

Complex Hilbert and classical-pair obstructions for $\mathbf{L}_{o,p}$

A partial answer to Question 3 of arXiv:1905.13552

Automated mathematical-discovery run `fa_banach_001`

13 August 2026

Status. Partial result, likely valid, subject to human review.

1 Source question and the scalar-field gap

For Banach spaces X, Y , the pair (X, Y) has property $\mathbf{L}_{o,p}$ if, for every $\varepsilon > 0$ and every fixed $x \in S_X$, there is $\eta(\varepsilon, x) > 0$ such that every $T \in S_{\mathcal{L}(X,Y)}$ satisfying $\|Tx\| > 1 - \eta(\varepsilon, x)$ has a point $x_0 \in S_X$ with

$$\|Tx_0\| = 1, \quad \|x_0 - x\| < \varepsilon.$$

On page 6, Dantas–Kim–Lee–Mazzitelli ask:

Question 3. Are there spaces X, Y with $\dim X, \dim Y \geq 2$ such that (X, Y) satisfies property $\mathbf{L}_{o,p}$?

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Then $\|T_n\| \leq 1$ and $\|T_n(x_2)\|_\infty = 1$, which implies $\|T_n\| = 1$. Since

$$\|T_n(x_1)\|_\infty = 1 - \frac{1}{n} > 1 - \eta(\varepsilon_0, x_1),$$

there is $z \in S_X$ such that $\|T_n(z)\|_\infty = 1$ and $\|z - x_1\| < \varepsilon_0$. So, since

$$1 = \|T_n(z)\|_\infty = \max \left\{ \left(1 - \frac{1}{n}\right) |x_1^*(z)|, |x_2^*(z)| \right\},$$

we have that $|x_2^*(z)| = 1$. Nevertheless, we have that $1 > \varepsilon_0 > \|z - x_1\| \geq |x_2^*(z) - x_2^*(x_1)| = |x_2^*(z)|$, which gives a contradiction. $\square$

Taking into account Propositions 2.6 and 2.7 and Corollary 2.12 below, we leave the following open question.

Question 3: Are there spaces X, Y with $\dim(X), \dim(Y) \geq 2$ such that (X, Y) satisfies property $\mathbf{L}_{o,p}$?

The source works over either $\mathbb{R}$ or $\mathbb{C}$. Its cited predecessor [2] proves a stronger fixed-point obstruction for every pair of *real* spaces of dimension at least two. That theorem already makes the answer negative over $\mathbb{R}$, but its decisive reduction and arc argument are explicitly real. The results below address the remaining complex territory.

2 Proof intuition

The Hilbert-domain construction turns the range geometry into a contraction from a Euclidean plane. The maximal-volume ellipsoid in a complex two-dimensional range ball is automatically circled, and John’s contact theorem provides two complex-independent points at which the contraction has norm one. A Hilbert-space diagonal map can fix one contact line while shrinking every vector transverse to it.

Composing the two maps leaves that line as the entire norming set, while a second, separated contact vector is only shrunk by a factor tending to one.

For $\ell_p^2 \rightarrow \ell_q^2$, a diagonal map gives the same mechanism when $p \leq q$. When $p > q$, the unperturbed identity embedding has a torus of maximizers $|z_1| = |z_2|$. A small symmetric off-diagonal perturbation breaks this degeneracy: its first variation on the torus is a positive multiple of the cosine of the relative phase, so exact maximizers select equal phases. The opposite-phase vector remains an almost maximizer because the perturbation tends to the identity.

3 Hilbert domains

Theorem 1 (Universal complex obstruction for Hilbert domains). *Let H be a complex Hilbert space and Y a complex Banach space, with $\dim H, \dim Y \geq 2$. Then (H, Y) fails property $\mathbf{L}_{o,p}$. More precisely, there are $x \in S_H$, $\delta > 0$, and $T_n \in S_{\mathcal{L}(H,Y)}$ such that*

$$\|T_n x\| \rightarrow 1, \quad \text{dist}(x, \{z \in S_H : \|T_n z\| = 1\}) \geq \delta \quad (n \geq 2).$$

Proof. Choose a complex two-dimensional subspace $Z \subset Y$. Let E be the John ellipsoid of the real four-dimensional centrally symmetric body B_Z . The John ellipsoid is unique. Since B_Z is invariant under multiplication by every scalar in $\mathbb{T}$, so is E . Consequently E is the unit ball of a Hermitian norm, and there is a complex-linear isomorphism $R : \ell_2^2 \rightarrow Z$ with

$$R(B_{\ell_2^2}) = E \subset B_Z, \quad \|R\| = 1.$$

The contact points $E \cap S_Z$ span Z over $\mathbb{R}$ by the John contact theorem [3, 4]. They therefore cannot lie in one complex line. Hence there are complex-linearly independent $u_1, u_2 \in S_{\ell_2^2}$ such that $\|Ru_1\| = \|Ru_2\| = 1$.

Identify ℓ_2^2 with a closed two-dimensional subspace H_0 of H , and let $Q : H \rightarrow H_0$ be the orthogonal projection. For $n \geq 2$, define the positive contraction A_n on H_0 by

$$A_n u_2 = u_2, \text{ and } A_n v = (1 - 1/n)v \quad (v \perp u_2),$$

and put $T_n = RA_n Q$. Since $\|T_n u_2\| = 1$, every T_n has norm one. Also $A_n u_1 \rightarrow u_1$, so $\|T_n u_1\| \rightarrow \|Ru_1\| = 1$.

If $z \in S_H$ and $\|T_n z\| = 1$, then

$$1 = \|RA_n Qz\| \leq \|A_n Qz\| \leq \|Qz\| \leq 1.$$

Equality in the last inequality forces $z \in H_0$, while equality for A_n forces $z \in \mathbb{C}u_2$. Thus the norming points of T_n are exactly $\mathbb{T}u_2$. Since u_1 and u_2 are complex-linearly independent, $\delta := \text{dist}(u_1, \mathbb{T}u_2) > 0$, proving the claim with $x = u_1$. $\square$

4 All complex two-dimensional classical pairs

We use the convention $2^{-1/\infty} = 1$.

Lemma 2 (Phase selection). *Let $1 \leq q < p \leq \infty$, set $a = 2^{-1/p}$, and define*

$$A_t(z_1, z_2) = (z_1 + tz_2, tz_1 + z_2) \quad (0 < t < 1).$$

If $t_j \downarrow 0$ and $z_j \in S_{\ell_p^2}$ satisfies $\|A_{t_j} z_j\|_q = \|A_{t_j}\|$, then every cluster point of (z_j) belongs to $\mathbb{T}(a, a)$.

Proof. Put

$$\Phi_t(z) = \|A_t z\|_q^q = |z_1 + tz_2|^q + |tz_1 + z_2|^q.$$

The maximum of Φ_0 on $S_{\ell_p^2}$ is $2a^q$, and equality holds exactly when $|z_1| = |z_2| = a$. (For $p = \infty$, this reads $|z_1| = |z_2| = 1$.) Compactness therefore shows that every cluster point z of (z_j) lies on this torus.

On a neighborhood of the torus both coordinates are bounded away from zero, so $(\Phi_t - \Phi_0)/t$ converges uniformly there to the one-sided derivative at $t = 0$. At $z = a(e^{i\alpha}, e^{i\beta})$ that derivative is

$$2qa^q \cos(\beta - \alpha).$$

This formula is also valid for $q = 1$, since neither coordinate vanishes. Let $z_+ = a(1, 1)$. Maximality of z_j gives

$$\begin{aligned} \frac{\Phi_{t_j}(z_j) - \Phi_0(z_j)}{t_j} &\geq \frac{\Phi_{t_j}(z_+) - \Phi_0(z_+)}{t_j} + \frac{\Phi_0(z_+) - \Phi_0(z_j)}{t_j} \\ &\geq \frac{\Phi_{t_j}(z_+) - \Phi_0(z_+)}{t_j}. \end{aligned}$$

Passing to the limit along a convergent subsequence forces $\cos(\beta - \alpha) \geq 1$. Thus the phases are equal and $z \in \mathbb{T}z_+$. $\square$

Theorem 3 (Every complex ℓ_p^2 - ℓ_q^2 pair fails). *For all $1 \leq p, q \leq \infty$, the complex pair (ℓ_p^2, ℓ_q^2) fails property $\mathbf{L}_{o,p}$.*

Proof. Suppose first that $p \leq q$. For $n \geq 2$ let

$$T_n(z_1, z_2) = ((1 - 1/n)z_1, z_2).$$

Since $\|w\|_q \leq \|w\|_p$, with the usual interpretation for $q = \infty$, $\|T_n\| = 1$. Moreover, $\|T_n e_1\| = 1 - 1/n \rightarrow 1$. Equality $\|T_n z\|_q = 1$ for $z \in S_{\ell_p^2}$ forces $|z_2| = 1$ (the first coordinate is strictly contracted), so every norming point remains at distance at least one from e_1 .

Now let $q < p$. With A_t as in Lemma 2, put $m_t = \|A_t\|_{\ell_p^2 \rightarrow \ell_q^2}$ and $T_t = A_t/m_t$. Let

$$x_- = a(1, -1), \quad x_+ = a(1, 1), \quad a = 2^{-1/p}.$$

The norm of the identity embedding is $c = 2^{1/q-1/p}$, so $m_t \rightarrow c$, while

$$\|A_t x_-\|_q = (1 - t)c.$$

Consequently $\|T_t x_-\|_q = (1 - t)c/m_t \rightarrow 1$.

Lemma 2 and compactness imply that the norming set of A_t (and hence of T_t) converges, in the upper Hausdorff sense, to $\mathbb{T}x_+$. Since $x_- \notin \mathbb{T}x_+$, $d := \text{dist}(x_-, \mathbb{T}x_+) > 0$. For all sufficiently small t , every norming point of T_t is at distance at least $d/2$ from x_- . A sequence $t \downarrow 0$ therefore violates $\mathbf{L}_{o,p}$ at the fixed point x_- . $\square$

5 Scope and upgrade attempts

Theorem 1 rules out every range once the domain is Hilbert; Theorem 3 rules out all classical two-dimensional pairs, including the non-Hilbert cross pairs not covered by the source's diagonal self-pair argument. These are negative subcases, not a full answer to Question 3.

Six focused routes were checked. First, the source and run indexes were audited for an existing answer. Second, citation tracing isolated the real theorem [2] and confirmed that its main result and proof are explicitly real. Third, the John-ellipsoid construction yielded Theorem 1. Fourth, the phase-selection

perturbation upgraded the classical-space search to Theorem 3. Fifth, a universal extension through a maximal-determinant contraction $X \rightarrow Y$ was pursued: it would suffice to produce two contact directions separated by a contractive one-sided perturbation, but nonsmooth common faces prevent the proposed perturbation from being certified. Sixth, quotient and Dvoretzky reductions were checked. Uniform property (P2) passes through the approximate-lifting argument in [2]; $\mathbf{L}_{o,p}$ does not, because its modulus depends on the fixed lift and the approximate lifts vary. No credible route past this obstruction remained.

A bounded search through 13 August 2026 used the exact question, the strings “ $\mathbf{L}_{o,p}$ ”, “property (P2)”, “operatorwise”, and “complex Banach spaces”, the source authors, and all eight OpenAlex citations of [2]. The 2022 expert overview [5] restates only the real universal theorem; no paper located states the complex extension or either theorem above. Novelty is plausible but provisional.

6 Verification notes

The Hilbert proof uses only equality in a chain of contractions after the standard John contact theorem. The $p \leq q$ proof is an exact diagonal calculation. The $p > q$ proof uses no numerical evidence: the only limiting step is uniform first-order expansion near a compact torus on which both coordinates are nonzero. This includes $q = 1$; compactness also includes $p = \infty$.

References

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- [2] S. Dantas, V. Kadets, S. K. Kim, H. J. Lee, and M. Martín, *There is no operatorwise version of the Bishop–Phelps–Bollobás property*, *Linear Multilinear Algebra* 68 (2020), 1767–1778; arXiv:1810.00684.
- [3] F. John, *Extremum problems with inequalities as subsidiary conditions*, in *Studies and Essays Presented to R. Courant*, Interscience, 1948, 187–204.
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